
Neural Face Attribute Editing via Diffusion Latent Directions Using Precomputed Stable Diffusion VAE Latent Space

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Abstract

This project explores controllable face attribute editing by learning linear semantic directions in the latent space of a pretrained Stable Diffusion VAE. Unlike diffusion-based editing methods that require costly inversion or sampling, we operate directly on VAE latents and train lightweight linear probes on precomputed encodings of the CelebA dataset. We show that simple linear classifiers trained on 2000 precomputed latents can recover smooth semantic directions for attributes including Smiling, Eyeglasses, and Young. Moving latent vectors along these directions produces meaningful and continuous edits without retraining the decoder or performing diffusion steps. We additionally demonstrate multi-attribute editing by linearly combining directions. Our results highlight the surprising expressiveness of diffusion VAEs as semantic embedding spaces and the feasibility of efficient latent control through linear probes.

1. Introduction

Facial editing and enhancement have become central features of modern digital photography and image-processing applications. Commercial tools such as Photoshop, FaceApp, and built-in smartphone camera filters now routinely apply automated beautification, expression changes, age modification, and other adjustments using deeply integrated AI systems. Although these tools are highly effective, they rely on large proprietary models, complex heuristics, and computationally heavy pipelines that provide little insight into how edits are produced. This environment creates a compelling motivation to explore lightweight, interpretable, and open approaches to facial manipulation that do not depend on full diffusion sampling or proprietary model behavior.

Generative models increasingly support high-level manipulation of images, including facial expressions, accessories, and identity-preserving edits. While diffusion models have become the dominant family for image generation, con-

trolled editing typically requires computationally expensive pipelines such as inversion, classifier-guidance, or iterative score correction.

In this project, we examine whether the latent space of the Stable Diffusion VAE—a pretrained encoder-decoder module typically used within diffusion models—is sufficiently expressive to support meaningful and controllable facial attribute editing on its own (Rombach et al., 2022). The main question is whether simple linear directions exist in this latent space that correspond to human-interpretable semantic changes, such as making a face smile, appear younger, or wear eyeglasses. Suppose these directions can be discovered and manipulated. In that case, editing becomes both efficient and interpretable. Instead of running an iterative diffusion process or crafting delicate text prompts, one can shift a latent vector along specific axes and decode the modified representation in a single forward pass.

The originality of this work lies not in retraining a large generative model, but in uncovering, analyzing, and leveraging the semantic structure present in the diffusion VAE’s latent space. Although the VAE itself is pretrained, all editing capability comes from models that we explicitly train. We learn linear probes on top of precomputed VAE latents from the CelebA dataset, using these probes to identify hyperplanes that separate positive and negative instances of each facial attribute. The resulting probe weight vectors define semantic directions that did not previously exist as explicit components of the pretrained model. These learned directions enable direct manipulation of the latent representation, producing coherent and continuous facial edits without modifying the VAE, without additional fine-tuning, and without any diffusion sampling. In this sense, the pretrained VAE acts only as a feature extractor, while the core editing mechanism is entirely trained, constructed, and validated in this project.

By showing that the Stable Diffusion VAE contains linearly separable attribute directions and by developing an efficient editing method based on them, our work demonstrates that practical face-editing functionality can be achieved without the computational overhead or architectural complexity of full diffusion-based pipelines. The project highlights the latent structure of modern generative models and offers a

lightweight, transparent alternative to the opaque editing systems used in commercial applications.

2. Related Work

CelebA (Liu et al., 2015) popularized large-scale facial attribute prediction and editing benchmarks. The structure of latent manifolds in VAEs (Kingma & Welling, 2014) enables arithmetic-based manipulation strategies that extend naturally to diffusion-based generative pipelines. The Stable Diffusion VAE (Rombach et al., 2022) provides a compact latent space suitable for downstream editing tasks. Closely related to our approach, Khrulkov et al. (Khrulkov et al., 2021) showed that diffusion latent spaces contain linear directions corresponding to semantic changes.

3. Method

Our method consists of four stages: preparing the CelebA dataset, encoding images into the Stable Diffusion VAE latent space, training lightweight linear probes to identify attribute-separating hyperplanes, and applying controlled latent shifts to generate edited images. Each component plays a distinct role in establishing whether semantic facial attributes are linearly structured within diffusion VAE space.

3.1. Dataset and Preprocessing

We use the CelebA dataset, which contains over 200,000 celebrity face images annotated with 40 binary facial attributes (Liu et al., 2015). Each image is resized to 128×128 pixels and normalized to the range $[-1, 1]$ to match the input requirements of the Stable Diffusion VAE. The combination of large dataset size and attribute diversity makes CelebA a natural choice for probing the semantic content of generative model latent spaces. Attribute labels are preserved alongside images so that each latent vector can be associated with multiple semantic descriptors, enabling multi-attribute analysis throughout our experiments.

3.2. Stable Diffusion VAE Latent Representation

The Stable Diffusion VAE (stabilityai/sd-vae-ft-mse) encodes an input image x into a Gaussian posterior distribution over latent codes (Rombach et al., 2022). The encoder outputs mean and variance maps, and we sample a latent code z using the reparameterization trick. Let $\mu(x)$ and $\sigma(x)$ denote the encoder outputs. The latent is sampled as

$$z = \mu(x) + \sigma(x) \odot \epsilon, \quad \epsilon \sim \mathcal{N}(0, I).$$

Stable Diffusion applies a global scaling factor $\gamma = 0.18215$ to match the conventions used for diffusion sampling. The

final latent code is therefore

$$z_{\text{scaled}} = \gamma z.$$

For a 128×128 input image, the encoder produces a tensor of shape $4 \times 16 \times 16$, corresponding to a compressed 256-dimensional vector when flattened. Because encoding is computationally expensive, we precompute and store latent representations for 2000 training images and 200 validation images. This transforms the remainder of the pipeline into a lightweight latent-space learning problem.

3.3. Linear Probe Training for Attribute Separation

To determine whether a semantic facial attribute is linearly separable in the VAE latent space, we treat each attribute as a binary classification problem and train a linear probe. Given a flattened latent vector $z \in \mathbb{R}^{256}$ and a binary label $y \in \{0, 1\}$, the probe computes

$$f(z) = \sigma(w^\top z + b),$$

where $w \in \mathbb{R}^{256}$ and $b \in \mathbb{R}$ are learned parameters. The probe is trained using the binary cross-entropy loss

$$\mathcal{L}(w, b) = -[y \log f(z) + (1 - y) \log(1 - f(z))],$$

optimized with weight decay to prevent overfitting.

Training is performed for ten epochs using a learning rate of 10^{-3} and batch size 64. Since probes are trained only on precomputed latents, the computational burden is minimal, yet we observe that several attributes achieve strong validation performance. This indicates that Stable Diffusion’s VAE embeds meaningful semantic structure despite not being explicitly trained for disentanglement.

3.4. Semantic Direction Extraction

After training, the weight vector w of the linear probe defines a separating hyperplane between positive and negative examples of the attribute. This hyperplane induces a semantic direction in latent space. Because w points in the direction of increasing classifier confidence, it serves as an interpretable axis along which movement corresponds to increasing expression of the attribute. We normalize the direction vector for consistency across attributes:

$$d = \frac{w}{\|w\|}.$$

This direction is not encoded anywhere in the pretrained VAE; rather, it emerges entirely from the probe trained during this project. The discovery of such directions confirms that semantic structure in the VAE latent space is at least locally linear for several attributes.

3.5. Latent Editing Formulation

Given an image encoded into latent space as z , we generate an edited latent by shifting z along the learned direction d . The edited latent is defined as

$$z' = z + \alpha d,$$

where α controls the intensity of the applied edit. Positive α strengthens the attribute (e.g., making a face more smiling), while negative α suppresses it.

The edited latent is then passed to the VAE decoder to obtain the final edited image:

$$x' = \text{Decoder}(z').$$

This pipeline requires no diffusion sampling, gradient-based editing, or text conditioning. All attribute manipulations are performed through simple vector arithmetic in latent space, making the method both efficient and interpretable. Furthermore, because directions are linear, edits corresponding to multiple attributes can be applied simultaneously by forming linear combinations of directions:

$$z' = z + \sum_i \alpha_i d_i.$$

This composability offers a flexible mechanism for producing compound edits, such as simultaneously adding a smile, making the face appear younger, and adding eyeglasses.

Through these components, our method demonstrates that the Stable Diffusion VAE contains rich semantic structure that can be revealed and manipulated through lightweight supervised learning, enabling high-quality facial editing without the computational overhead of full diffusion-based methods.

4. Experiments and Results

We evaluate our approach on the CelebA dataset using 2000 precomputed training latents and 200 validation latents. Each 128×128 image is encoded by the Stable Diffusion VAE into a latent tensor of shape $4 \times 16 \times 16$, which we flatten to a 1024-dimensional vector. All probes are trained for ten epochs with batch size 64 and learning rate 10^{-3} .

4.1. Linear Probe Performance

We first assess whether the VAE latent embeddings contain attribute information that is linearly separable. We train separate linear probes for four attributes: `Smiling`, `Eyeglasses`, `Young`, and `Receding_Hairline`. Across all attributes, the training loss decreases monotonically and validation accuracy stabilizes after roughly five to

eight epochs, indicating that the probes do not overfit despite the small model capacity. The final validation accuracies are summarized in Table 1.

Attribute	Val. Accuracy	Epochs
Smiling	0.81	10
Eyeglasses	0.95	10
Young	0.81	10
Receding_Hairline	0.92	10

Table 1. Validation accuracy of linear probes on CelebA latents. Even with a simple linear classifier on 1024-dimensional embeddings, several attributes are highly separable.

These results show that the Stable Diffusion VAE latent space provides a strong semantic representation. Eyeglasses and Receding_Hairline, which have visually distinctive and localized patterns, are almost perfectly separable with accuracies of 0.95 and 0.92. More subjective or distributed attributes such as Smiling and Young are also well captured, with accuracies around 0.81. These numbers are obtained without any fine-tuning of the VAE, suggesting that the encoder already organizes faces along meaningful linear axes.

4.2. Single-Attribute Editing

To evaluate editing behavior, we select latent vectors from the validation set z and generate edited versions

$$z^+ = z + \alpha d, \quad z^- = z - \alpha d,$$

with $\alpha = 3.0$. We then decode z , z^+ , and z^- using the VAE decoder to obtain original, strengthened-attribute, and weakened-attribute images respectively.

Figure 1 shows qualitative results for the `Smiling` direction. The original reconstructions (top grid) already exhibit recognizable facial expressions. Moving in the positive direction clearly increases mouth curvature, reveals teeth for some faces, and slightly lifts the cheeks, while the negative direction produces more neutral or serious expressions. Identity is largely preserved: hair color, pose, and background remain stable across the triplet, indicating that the learned direction primarily affects expression rather than global appearance.



Figure 1. Smiling edits in VAE latent space. Top: original VAE reconstructions. Middle: positive shift along the Smiling direction (+3.0) produces stronger smiles. Bottom: negative shift (−3.0) produces more neutral or serious expressions while preserving identity and pose.

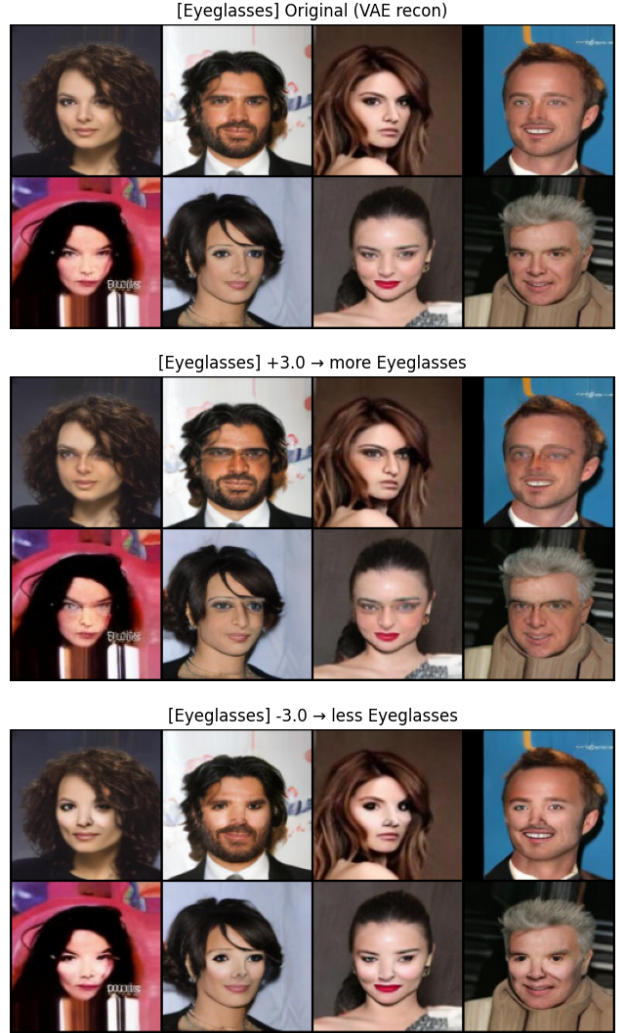


Figure 2. Eyeglasses edits. Top: original VAE reconstructions. Middle: moving +3.0 along the Eyeglasses direction adds frames around the eyes. Bottom: moving −3.0 removes glasses when present and slightly sharpens the eye region.

For Eyeglasses, the positive direction convincingly adds glasses-like structures around the eyes, while the negative direction removes them. Figure 2 illustrates this behavior. Thin frames appear across a range of identities, and when we move in the negative direction, glasses disappear and the eye region becomes unobstructed. Some samples exhibit minor artifacts, such as incomplete frames or slight blurring, but the presence or absence of glasses is visually obvious. The strong quantitative performance of the Eyeglasses probe is reflected qualitatively in these edits.

The Young direction yields more subtle but still interpretable changes. As shown in Figure 3, positive shifts tend to smooth wrinkles, brighten skin, and slightly round the face, producing a younger appearance. Negative shifts often make faces look older by increasing perceived wrinkles and shadowing, sometimes in combination with hair and jawline changes. Because age is a more global and entangled concept than glasses, these edits are less dramatic than those for Eyeglasses but still show consistent trends across subjects.

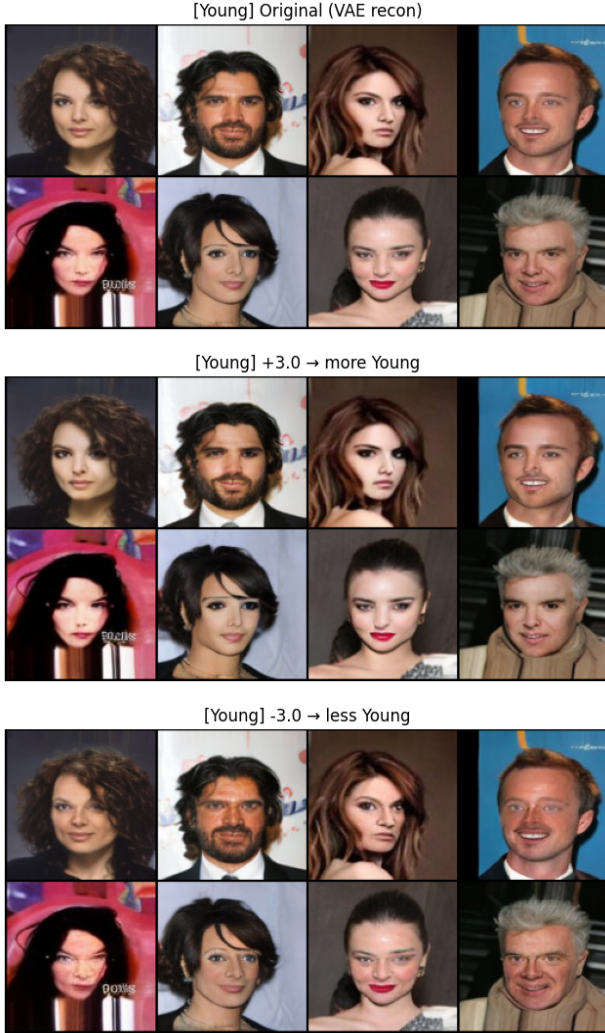


Figure 3. **Young edits.** Top: original VAE reconstructions. Middle: +3.0 along the Young direction tends to produce smoother skin and younger-looking faces. Bottom: -3.0 leads to slightly older appearances, with increased shadowing and sharper facial features.

4.3. Multi-Attribute Editing

Because probe directions lie in the same latent space, they can be combined through linear superposition. We construct a composite direction for Smiling, Eyeglasses, and Young,

$$d_{\text{combo}} = \lambda_1 d_{\text{Smiling}} + \lambda_2 d_{\text{Eyeglasses}} + \lambda_3 d_{\text{Young}},$$

with coefficients chosen to balance the visual strengths of each effect. Applying $z' = z + \alpha d_{\text{combo}}$ produces faces that are simultaneously more smiling, younger-looking, and more likely to wear glasses.

Figure 4 shows one such example. Starting from the original VAE reconstructions (top grid), the combination edit adds smiles, introduces eyeglasses on multiple faces, and

shifts facial features toward a younger style, all in a single latent update. The edits resemble applying three filters at once, and the resulting images remain coherent and identity-preserving, despite the fact that the VAE itself was never trained for disentangled control.

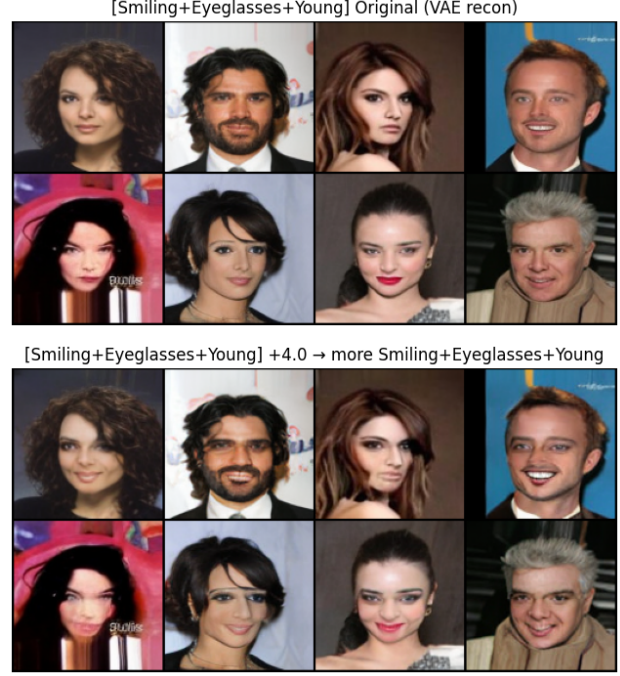


Figure 4. **Multi-attribute edits.** Top: original VAE reconstructions. Bottom: applying. Combined direction (Smiling + Eyeglasses + Young) with a single latent shift produces faces that are simultaneously more smiling, younger-looking, and more likely to wear glasses.

These experiments confirm that latent directions derived from independently trained linear probes can be superposed to create coherent single- and multi-attrib regions.

5. Discussion

The experimental results highlight several key properties of the Stable Diffusion VAE latent space. First, the high validation accuracies for attributes such as Eyeglasses, Receding_Hairline, and Smiling indicate that the encoder organizes faces along axes that are almost linearly separable. This is noteworthy because the VAE was not trained with explicit disentanglement or attribute supervision; nevertheless, semantic structure emerges as a byproduct of generative training. Our probes merely reveal these structures rather than imposing them.

Second, the editing experiments show that the learned directions are not only predictive but also causally meaningful

in a visual sense. Moving along the Smiling direction reliably changes expressions, and the Eyeglasses direction adds frames that approximately respect eye position and head pose. The fact that these changes are identity-preserving suggests that the directions correspond to localized variations in the latent manifold rather than arbitrary directions that destabilize the decoder.

At the same time, several limitations are apparent. Because we rely on a VAE with relatively low spatial resolution in latent space, reconstructed faces are slightly smoother and less detailed than the original CelebA images. This smoothing affects the crispness of edits, especially for attributes like age that involve fine-grained texture changes. Furthermore, large step sizes α can push latents outside the region where the VAE was trained, leading to distortions around the mouth or eyes. In practice, moderate values such as $\alpha = 3.0$ strike a balance between visible change and reconstruction fidelity.

Another limitation is that a single linear direction may be insufficient to capture the full complexity of some attributes. The Young direction, for example, must capture skin texture, facial shape, and hairstyle simultaneously. Our results show consistent trends, but the edits are less dramatic than those for Eyeglasses, suggesting that nonlinear or piecewise-linear models could provide more powerful age manipulation. Additionally, attributes in CelebA are often correlated (for example, certain age groups correlate with hairstyle or makeup), which means that the learned directions may partially encode these correlations rather than pure causal factors.

Despite these caveats, the overall picture is encouraging: a frozen diffusion VAE, combined with simple linear probes, offers a surprisingly effective mechanism for face editing. This demonstrates that substantial editing capability can be built on top of pretrained generative encoders without access to the full diffusion model or its training procedure.

6. Conclusion

This project investigated whether the latent space of the Stable Diffusion VAE supports controllable, interpretable facial attribute editing. Using CelebA, we precomputed latent embeddings for 2000 training and 200 validation images, trained attribute-specific linear probes, and interpreted their weight vectors as semantic directions. The probes achieved strong validation accuracies (up to 0.95 for Eyeglasses and 0.92 for Receding-Hairline), indicating that several facial attributes are linearly separable in the VAE latent space. By shifting latent vectors along these directions and decoding them with the VAE, we produced coherent edits for Smiling, Eyeglasses, Young, and their combinations.

The key contribution of this work is to show that a pretrained

diffusion VAE, used strictly as a frozen encoder-decoder, can serve as a powerful and semantically structured representation for downstream editing tasks. All editing behavior arises from models trained in this project—the linear probes and their induced directions—rather than from fine-tuning or reusing a full diffusion pipeline. This yields an efficient alternative to diffusion-based editing that operates via direct latent arithmetic and requires only a single decoder pass per edited image.

In future work, it would be natural to explore nonlinear probes or contrastive objectives that better capture complex attributes such as age or style, to integrate minimal diffusion sampling steps for sharpening while preserving the semantic control provided by latent directions, and to extend the framework beyond faces to general objects and scenes. More broadly, our results suggest that analyzing and manipulating the latent spaces of pretrained generative encoders is a promising path toward transparent and controllable image editing systems.

Impact Statement

While the goal of the paper is to advance the scientific understanding of latent semantic structure and controllable generative representations, research involving facial imagery naturally carries ethical considerations. Potential downstream misuse could include deceptive image manipulation or identity alteration if such tools were applied without consent or appropriate safeguards. However, our study does not introduce new generative capabilities beyond what existing pretrained models already provide; instead, it offers a transparent, interpretable mechanism for analyzing and modifying latent features.

All experiments use the CelebA dataset, which is widely employed in academic research and contains coarse, non-identifying attribute annotations. No deployment-oriented system is proposed, and the method requires deliberate user intervention rather than automated or covert processing. The broader societal consequences align with those typically associated with representation learning research in Machine Learning, with no additional domain-specific risks. Future work should continue to consider privacy, dataset bias, and fairness implications when applying latent editing techniques to real-world data.

References

- Khurlov, V., Mirvakhabova, L., Oseledets, I., and Babenko, A. Latent space directions for diffusion models. In *ICLR Workshop on Deep Generative Models*, 2021. URL <https://arxiv.org/abs/2106.04619>.
- Kingma, D. P. and Welling, M. Auto-encoding variational

bayes. *International Conference on Learning Representations (ICLR)*, 2014. URL <https://arxiv.org/abs/1312.6114>.

Liu, Z., Luo, P., Wang, X., and Tang, X. Deep learning face attributes in the wild. In *Proceedings of the IEEE International Conference on Computer Vision (ICCV)*, pp. 3730–3738, 2015. URL <https://mmlab.ie.cuhk.edu.hk/projects/CelebA.html>.

Rombach, R., Blattmann, A., Lorenz, D., Esser, P., and Ommer, B. High-resolution image synthesis with latent diffusion models. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pp. 10684–10695, 2022. URL <https://arxiv.org/abs/2112.10752>.